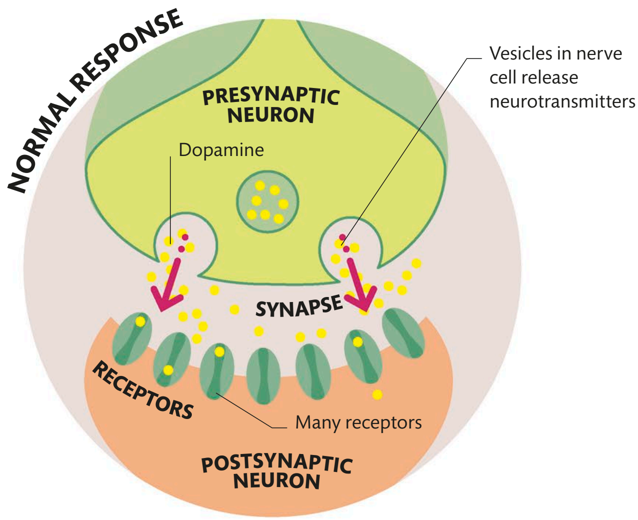
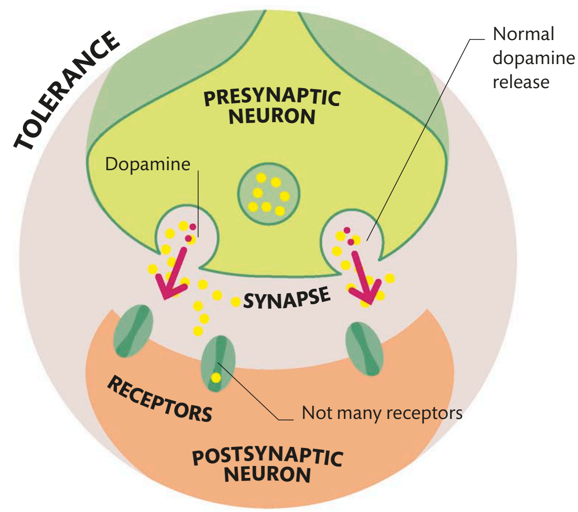
Addiction

Most drugs of abuse cause huge amounts of dopamine to build up in the reward system—far more than natural rewards like food or sex. This creates a powerful drive to seek out more of the drug. It also causes the brain to reduce the number of dopamine receptors, so natural rewards no longer give the same sensation. This means the user loses the urge to seek out things like food and social engagement. Instead, drug cues become powerful triggers for dopamine release, causing intense cravings, even when the user consciously wants to stop and no longer enjoys the drug.

UP TO 60% OF ADDICTION RISK STEMS FROM GENETIC FACTORS



Flooded with dopamine
Some drugs of abuse increase dopamine release, while others stop it from being recycled. The buildup in the synapse produces a large response in the brain, triggering the drive to seek out more of the drug. Environmental cues become linked with the drug and can trigger cravings in the future.



Under tolerance
Over time, the brain reduces the number of dopamine receptors to counteract the excess. Now, when normal amounts of dopamine are released, they have little effect. The user may need bigger and bigger doses of the drug to feel its effect, and their desire for other rewards decreases.

WHY IS JUNK FOOD SO TASTY?

Most junk food contains lots of sugar, salt, and fat, which trigger our reward system. This would have helped us survive when food was scarce.

WANTING VERSUS LIKING

The reward pathway is often called a "pleasure pathway," and dopamine a "pleasure chemical," but this is not accurate. Dopamine in the nucleus accumbens drives "wanting" of a reward, but it is common for addicts to experience strong cravings without liking the effects of the drug. Pleasure is likely to be caused by other neurotransmitters such as opioids or endocannabinoids.